

|                                    |                                                                                                                                                                                                                                                         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title                      | End-to-End Machine Learning Product for Telecom Customer Churn Prediction                                                                                                                                                                               |
| Skills take away From This Project | Python, Pandas, NumPy, Data Cleaning, Exploratory Data Analysis, Feature Engineering, Statistics, Machine Learning, Class Imbalance Handling, Model Evaluation, Explainable AI (SHAP), REST API Development, Flask/FastAPI, Git & GitHub, Documentation |
| Domain                             |                                                                                                                                                                                                                                                         |

### Problem Statement:

You are working as a **Data Scientist** for a telecom company.

The company is facing **high customer churn**, which negatively impacts revenue and growth.

You are given historical customer data containing:

- Demographics
- Account details
- Services subscribed
- Billing information

Your task is to build an **end-to-end Machine Learning product** that:

- Predicts whether a customer will churn
- Explains *why* the customer is likely to churn
- Helps the business take proactive retention actions
- Exposes predictions via a production-ready API

## Business Use Cases

- Identify customers at **high risk of churn**
- Reduce customer acquisition cost
- Improve customer lifetime value (CLV)
- Enable personalized retention offers
- Support management decisions with data-driven insights

## Approach

### 3.1 Data Understanding & Ingestion

Load the telecom customer churn dataset and understand the business meaning of each column.

#### Dataset Columns

- customerID
- gender
- SeniorCitizen
- Partner
- Dependents
- tenure
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges
- Churn (**Target**)

## 3.2 Exploratory Data Analysis (EDA)

Perform detailed analysis to answer key business questions:

- Do **senior citizens** churn more?
- How does **tenure** affect churn?
- Which **contract types** have the highest churn?
- Does **monthly billing amount** influence churn?
- Which services are associated with higher churn?

### Mandatory Visualizations

- Churn distribution (class imbalance)
- Tenure vs churn
- MonthlyCharges vs churn
- Contract vs churn
- InternetService vs churn
- Correlation heatmap

## 3.3 Data Cleaning & Preprocessing

- Convert `TotalCharges` to numeric (handle missing values)
- Drop non-informative columns (`customerID`)
- Handle missing and inconsistent values
- Encode categorical variables
- Scale numerical features (`tenure`, `MonthlyCharges`, `TotalCharges`)
- Split data into Train / Validation / Test sets

## 3.4 Feature Engineering

Create meaningful features such as:

- $\text{AvgMonthlySpend} = \text{TotalCharges} / \text{tenure}$
- Tenure buckets (New / Medium / Loyal)
- Service count per customer
- Contract duration flags
- High-value customer indicator

**Important:** Each engineered feature must have business meaning.

### 3.5 Handling Class Imbalance

Since churn is imbalanced:

- Use class weights or resampling techniques
- Avoid accuracy-only evaluation
- Focus on Recall, Precision, F1-score, ROC-AUC

### 3.6 Model Building & Evaluation

Train and compare multiple models:

- Logistic Regression (baseline)
- Random Forest
- Gradient Boosting / XGBoost (Optional)

Evaluation Metrics

- ROC-AUC
- Precision
- Recall
- F1-score
- Confusion Matrix

### 3.7 Threshold Optimization

- Default threshold (0.5) is not mandatory
- Choose optimal threshold based on:
  - Cost of losing a customer
  - Cost of retention incentives

### 3.8 Model Explainability

Apply Explainable AI techniques:

- SHAP global feature importance
- SHAP local explanations per customer

Explain:

- Why customers churn
- Which features drive churn the most

### 3.9 Deployment (ML Product Layer)

Build a REST API using Flask or FastAPI.

Input Example

```
{  
  "gender": "Female",  
  "SeniorCitizen": 0,  
  "Partner": "Yes",  
  "Dependents": "No",  
  "tenure": 5,  
  "PhoneService": "Yes",  
  "InternetService": "Fiber optic",  
  "Contract": "Month-to-month",  
  "MonthlyCharges": 89.85  
}
```

Output Example

```
{  
  "churn_probability": 0.81,  
  "churn_risk": "High"  
}
```

## Technologies and Tools

| Technology                 | Purpose          |
|----------------------------|------------------|
| <b>Python</b>              | Core programming |
| <b>Pandas, NumPy</b>       | Data processing  |
| <b>Matplotlib, Seaborn</b> | Visualization    |
| <b>Scikit-learn</b>        | ML modeling      |
| <b>SHAP</b>                | Explainability   |
| <b>Flask / FastAPI</b>     | API deployment   |
| <b>Git &amp; GitHub</b>    | Version control  |
| <b>Docker (Optional)</b>   | Containerization |

## Results

Each student must submit:

Structured folders:

data/

notebooks/

src/

models/

api/

utils/

## Notebooks

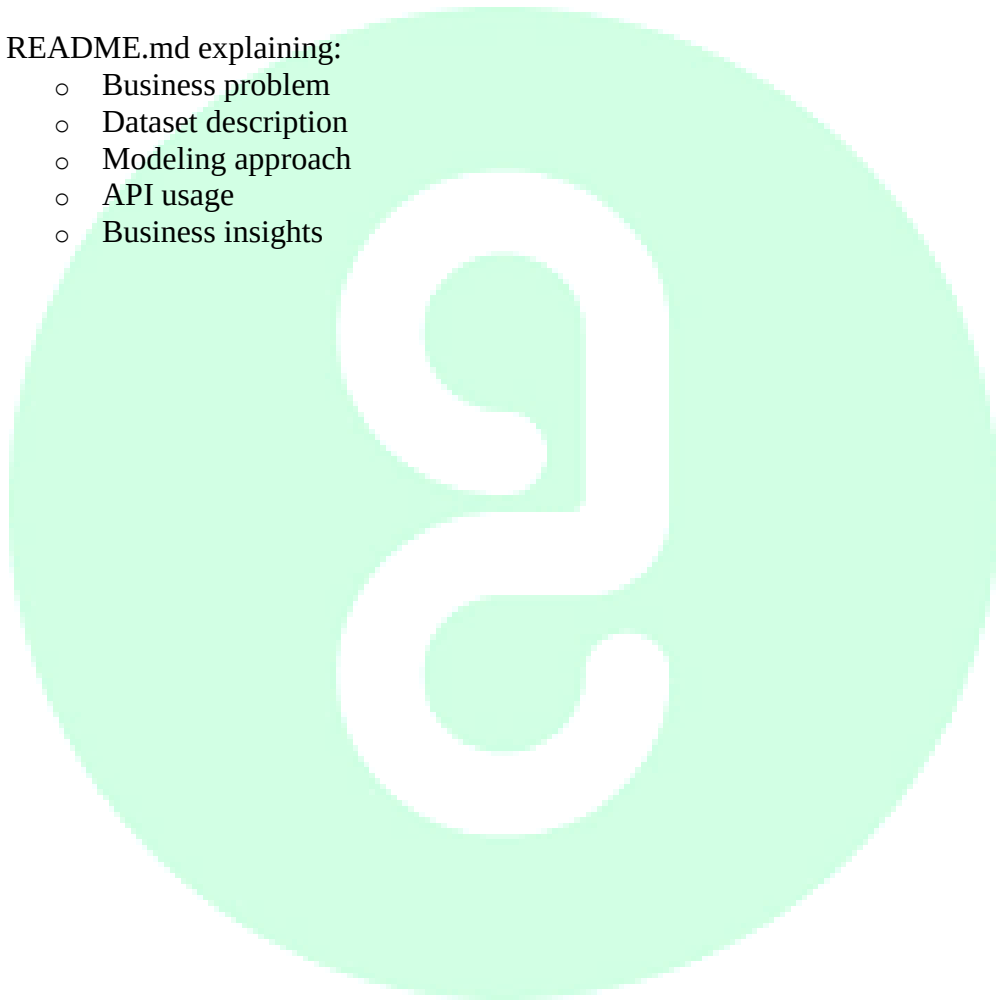
- EDA Notebook
- Modeling Notebook
- Explainability Notebook

## API

- `/predict` endpoint
- JSON input/output
- Input validation

## Documentation

- README.md explaining:
  - Business problem
  - Dataset description
  - Modeling approach
  - API usage
  - Business insights



## Project Evaluation Metrics

| Evaluation Criteria         | Marks      |
|-----------------------------|------------|
| Data Understanding & EDA    | 15         |
| Feature Engineering         | 10         |
| Model Building & Evaluation | 20         |
| Class Imbalance Handling    | 10         |
| Threshold Optimization      | 10         |
| Explainability (SHAP)       | 15         |
| API Deployment              | 10         |
| Documentation               | 5          |
| Code Quality                | 5          |
| <b>Total</b>                | <b>100</b> |

### Bonus (+5):

- Dockerized API
- Streamlit dashboard
- Business cost analysis

## Dataset Explanation

### 1. Customer Demographics

- gender
- SeniorCitizen
- Partner
- Dependents

### 2. Account & Services

- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies



### 3. Billing & Contract

- Contract
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges
- tenure

### 4. Target Variable

- Churn (Yes / No)

## Project Deliverables

- Private GitHub Repository
- Clean notebooks
- Model artifacts
- REST API
- README.md
- Final business summary

### Project Guidelines:

1. **GitHub Repository**
  - Push your final project to your GitHub account.
  - Repository must be private during the assignment period.
  - Add the instructor's GitHub ID as a collaborator for review.
2. **Naming Convention**
  - GitHub Repo Name: `telecom-churn-ml-{yourname}`
  - Example: `telecom-churn-ml-amit`

### Timeline:

Define the project timeline, including milestones and deadlines.

## PROJECT DOUBT CLARIFICATION SESSION ( PROJECT AND CLASS DOUBTS)

**About Session:** The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

**Note: Book the slot at least before 12:00 Pm on the same day**

**Timing:** Monday-Saturday (4:00PM to 5:00PM)

**Booking link :** <https://forms.gle/XC553oSbMJ2Gcfug9>

**For DE/BADM project/class topic doubt slot clarification session:**

**Booking link :** <https://forms.gle/NtkQ4UV9cBV7Ac3C8>

**Session timing:**

**For DE: 04:00 pm to 5:00 pm every saturday**

**For BADM 05:00 to 07:00 pm every saturday**

## LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

**About Session:** The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

**Note: This form will Open only on Saturday (after 2 PM ) and Sunday on Every Week**

**Timing:**

**For BADM and DE**

**Monday-Saturday (11:30AM to 1:00PM)**

**For DS and AIML**

**Monday-Saturday (05:30PM to 07:00PM)**

**Booking link :** <https://forms.gle/1m2Gsro41fLtZurRA>